

Bandwidth Estimation using Utility Maximization

Bruce Spang
UMass Amherst
bspang@cs.umass.edu

Ramesh Sitaraman
UMass Amherst
ramesh@cs.umass.edu

Donald Towsley
UMass Amherst
towsley@cs.umass.edu

1. INTRODUCTION

In networking, we often try to divine details of internal properties of networks from the end-to-end behavior of traffic. In the area of network tomography, exemplified by [2] and [4], the goal is to learn link properties such as queueing delay or packet loss. Work on bandwidth estimation, such as [3] and [5], has as its goal to learn the maximum capacity along a path between source and destination. In both of these areas, the general approach is to send pairs of packets or a “train” of many packets and observe how the network affects the returned acknowledgments. For instance, [3] shows how to infer the capacity of a path by using the inter-packet spacing of acknowledgments to these packet pairs.

In this paper, we propose a new approach to network tomography using Kelly et al.’s optimization-based models of TCP [6]. In [6], Kelly et al. show that TCP implicitly adjusts the sending rate of flows to solve an optimization problem. We ask: what can we learn about internal network characteristics, *given the equilibrium rates for the flows that we can observe*?

We summarize our contributions below. In Section 2, we formally define our network model that derives from the work of Kelly et al. [6]. Using this model, we show the following new results:

Path Capacity In Section 3, we show a new method for estimating the capacity of a single bottleneck. This is the same setting as existing bandwidth estimation work, such as [5].

Per-link Price In Section 4, we show how to estimate per-link prices in our model, which are equivalent to loss probabilities or average queueing delays. This is a classic problem in network tomography, as described by [2].

Network Capacity In an arbitrary network, flows need not be constrained by the minimum capacity link along their path, and instead may be limited by a larger link with more competition. In Section 5, we look at the new problem of estimating capacities in an arbitrary network. We give a set of equations to find bottleneck capacities even in the presence of background flows.

2. MODEL

We model a network using the utility maximization framework pioneered by Kelly et al. [6]. Let L denote the set of links in the network and A the set of all flows in the network. Each flow $a \in A$ traverses a certain set of links. Let

the *routing matrix* $R : |L| \times |A|$ be the matrix representing which flows traverse which links: $R[l, a] = 1$ if flow a uses link l , and $R[l, a] = 0$ otherwise. Let C be the vector of link capacities where C_l is the capacity of link l . Let x be a vector of rates for all the flows $a \in A$ where x_a is the rate of a single flow $a \in A$. Since we cannot exceed the capacity of each link, we have the constraint $Rx \leq C$.

We assume that each flow $a \in A$ uses a congestion-control algorithm that can be implicitly described as optimizing a concave, strictly increasing utility function $U_a(x_a) : \mathbb{R}^+ \rightarrow \mathbb{R}$.

A user performing network tomography is interested in learning network parameters, such as the link capacities C_l . We assume that such a user can add or remove some subset of flows from the network, and observe the equilibrium rates of those flows. We call the flows controlled by this user *foreground flows* and let F denote the set of foreground flows. In a network, there may also be flows that this user cannot control or observe. We call these the *background flows* and let B denote the set of background flows. We assume F and B are disjoint and $A = F \cup B$.

Following Kelly et al. [6], we model the network as solving the following optimization problem in equilibrium,

$$\begin{aligned} \max_x \quad & \sum_{f \in F} U_f(x_f) + \sum_{b \in B} U_b(x_b), \\ \text{subject to} \quad & Rx \leq C, \\ & x \geq 0. \end{aligned} \quad (1)$$

Let λ be the dual variables of Problem (1). We also refer to λ as the *link prices* in the style of [6]. The Lagrangian dual of Problem (1) is given by

$$\min_{\lambda} \sup_{x \geq 0} \sum_{f \in F} U_f(x_f) + \sum_{b \in B} U_b(x_b) + \lambda^T (Rx - C) \quad (2)$$

A major result by Kelly et al. [6] shows that rate controlled flows and in particular TCP flows, have equilibrium rates which solve Problem (1). Furthermore, if the TCP flows respond to loss signals (e.g. in TCP Reno or TCP Cubic) the dual variables λ from Problem (2) of the optimization problem correspond to the loss probabilities at each link when loss is small. If the TCP flows respond to queueing delays (e.g. TCP Vegas), the dual variables of the optimization problem correspond to per-link average queueing delays. For more details, see [6] or ([7], Sec. 7.1).

Our general approach to solving tomography problems is to assume that the network is at equilibrium and therefore

the rates of each flow are a solution to Problem (1). To infer network details, we often rely on the KKT Theorem ([1], Sec. 5.5.3): because Problem (1) is concave, a vector of flow rates x and link prices λ satisfy the following KKT conditions if and only if they are the optimal solution to Problem (1):

$$Rx \leq C, \quad (3)$$

$$x \geq 0, \quad (4)$$

$$\lambda \geq 0, \quad (5)$$

$$(Rx - C)^T \lambda = 0, \quad (6)$$

$$\nabla U(x) = R^T \lambda. \quad (7)$$

3. SINGLE BOTTLENECK

In this section, we consider the case when there can only be one active constraint in the network. This corresponds to when either the network has a single link, the network has many links but only one bottleneck, or when there are back-to-back bottlenecks with the same capacity.

We assume that we know a-priori the set of possible congestion control algorithms used by background flows. For example, we might assume that background traffic will use one of the default algorithms for Linux, Mac OS X, or Windows, which are TCP Cubic and Compound TCP.

Our measurement algorithm runs flows of each of these congestion control algorithms and observe their behavior.

Let K be the set of congestion control algorithms, and let $U_k(x)$ be the utility function corresponding to congestion control algorithm k . We assume $U_k(x)$ is strictly concave and increasing for all algorithms k . Let n_k denote the known number of foreground flows of type k and m_k the unknown number of background flows of type k . Let x_k be the equilibrium rate achieved by one foreground flow of type k and y_k be the equilibrium rate for one background flow of type k . This corresponds to the following utility maximization model,

$$\begin{aligned} & \max_{x, y} \sum_{k \in K} n_k U_k(x_k) + m_k U_k(y_k), \\ & \text{subject to } \sum_{k \in K} n_k x_k + m_k y_k \leq C, \\ & x_k, y_k \geq 0 \quad \forall k \in K. \end{aligned}$$

PROPOSITION 1. *Under the previous assumptions, capacity C , and the number of each type of background flow can be found by solving the following matrix equation:*

$$\begin{bmatrix} x_1^{(1)} & \dots & x_{|K|}^{(1)} & -1 \\ x_1^{(2)} & \dots & x_{|K|}^{(2)} & -1 \\ & \ddots & & \\ x_1^{(n)} & \dots & x_{|K|}^{(n)} & -1 \end{bmatrix} \begin{bmatrix} m_1 \\ \vdots \\ m_{|K|} \\ C \end{bmatrix} = \begin{bmatrix} -\sum_{k \in K} n_k^{(1)} x_k^{(1)} \\ -\sum_{k \in K} n_k^{(2)} x_k^{(2)} \\ \vdots \\ -\sum_{k \in K} n_k^{(n)} x_k^{(n)} \end{bmatrix} \quad (8)$$

PROOF. The proof relies on the foreground flows of type k observing the same equilibrium rates as background flows of type k .

Let x_k^* denote the equilibrium rate of foreground flows of type k , and define y_k^* similarly. From KKT Condition (7),

we have for all k ,

$$n_k U'_k(x_k^*) = n_k \lambda^*,$$

$$m_k U'_k(y_k^*) = m_k \lambda^*.$$

Thus for all k , $U'_k(x_k^*) = U'_k(y_k^*) = \lambda^*$. Since all utility functions are strictly concave and increasing, their derivative is strictly decreasing. Therefore for all k , $x_k^* = y_k^*$.

Since there is one constraint, by KKT Condition (6),

$$\sum_{k \in K} n_k x_k + m_k y_k = C.$$

Substituting x_k for y_k by the above and rearranging terms yields

$$-\sum_{k \in K} n_k x_k = m_k x_k - C.$$

Making many such measurements gives a system of these equations, can be written in the form of (8). In numerical experiments, this matrix has always had full rank when $n = |K|$ and we have been able to find all m_k and C . Unfortunately, we have not been able to prove conditions under which the matrix in (8) has full rank. $\square$

4. ESTIMATING PRICES

A classic problem in network tomography is to find the per-link loss and queueing delay using end-to-end measurements. In this section, we show how to solve this problem using our models.

In our model, this problem relates to inferring the dual variables λ from Problem (2) from measurements of the foreground flows. When the congestion control algorithm uses delay as its congestion signal, the dual variables correspond to average queueing delays ([7], Sec. 7.1.3). When the congestion controllers use loss signals and loss probabilities are small, the dual variables are approximated by the per-link loss probabilities ([7], Sec 2.4.1). Knowing these dual variables is an interesting problem in and of itself, and will also be useful in Section 5.

Like existing tomography work, we need a condition to find per-link prices from the observable behavior of foreground flows. Let K be the subset of links which are used by some foreground flow. Let R_F be the routing matrix for the foreground flows, defined as $R_F[l, f] = 1$ if flow $f \in F$ crosses link $l \in K$ and $R_F[l, f] = 0$ otherwise.

We say that the set of links K is *identifiable* given a set of foreground flows F if R_F has rank $|K|$.

PROPOSITION 2. *Let R_F be an identifiable foreground routing matrix as defined above, x_F the optimal set of foreground rates, and $\nabla_F U(x_F)$ the gradient of the foreground flows. The unique vector λ_K of link prices for links in K is given by solving the matrix equation*

$$\nabla_F U(x_F) = R_F^T \lambda_K. \quad (9)$$

PROOF. Equation (9) has a unique solution by our assumption that R_F is identifiable.

By the KKT condition (7), at equilibrium the optimal set of rates x and link prices λ satisfy the condition

$$\nabla U(x) = R^T \lambda.$$

This condition is equivalent to the condition that for all flows $a \in F \cup B$, $U'_a(x_a) = \sum_{l \in a} \lambda_l$. Considering the conditions associated with links $l \in K$ as a matrix yields (9). $\square$

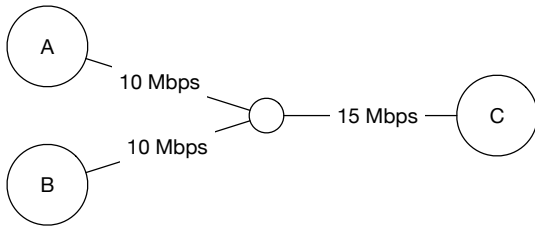


Figure 1: A three node network where the 15 Mbps link, not the link with minimum capacity, will constrain TCP's throughput

5. CAPACITIES OF A NETWORK

In this section, we use knowledge of the foreground routing matrix to simultaneously find the capacities of bottleneck links. Note that bottlenecks found by considering each path as a single bottleneck using Section 3 may not accurately describe the network. For an example, consider the three node network shown in Figure 1. Considering either path A to C or B to C on its own, the bottleneck is the 10 Mbps link. However, if the two paths split the network fairly, they will be constrained to 7.5 Mbps by the 15 Mbps link.

We assume that all flows in the network use a weighted log utility function, that is for all flows a , $U_a(x) = w_a \log x$. We assume the foreground routing matrix R_F is known and *identifiable* (as defined in Section 4). We do not assume that the weights or routes for background traffic are known, but we do assume that if some link is a bottleneck for a background flow, it is also a bottleneck for a foreground flow. This corresponds to the following model,

$$\begin{aligned} \max_x \quad & \sum_{f \in F} U_f(x_f) + \sum_{b \in B} U_b(x_b), \\ \text{subject to} \quad & Rx \leq C, \\ & x \geq 0. \end{aligned} \quad (10)$$

Using this model, we have the following relationship between the weights of all the flows and the prices and capacities at each link. We can iteratively use this relationship to find C and $\sum_{b \in B} w_b$.

LEMMA 3. *If x are the optimal primal variables and λ the optimal dual variables from Problem (10), then*

$$\sum_{a \in F \cup B} w_a = \lambda^T C \quad (11)$$

PROOF. Expanding KKT condition (6) yields $\lambda^T(Rx) = \lambda^T C$. The dot product is commutative, so $(Rx)^T \lambda = \lambda^T C$. Expanding the left-hand side yields $x^T R^T \lambda = \lambda^T C$. Substituting $R^T \lambda$ for $\nabla U(x)$ using KKT condition (7) yields $x^T \nabla U(x) = \lambda^T C$. Under the assumption that all flows use weighted-log utility functions, $x^T \nabla U(x) = \sum_{a \in F \cup B} w_a$. Substituting this yields the desired result. $\square$

Let $W_F = \sum_{f \in F} w_f$, which is known. For the background flows, let $W_B = \sum_{b \in B} w_b$, which is unknown. By the assumption that R_F is identifiable, we can find the vector of prices λ for all foreground links by solving (9). Since we have assumed the background bottlenecks are a subset of the foreground bottlenecks, (9) allows us to find the prices of all bottlenecks in the network. This is still not enough

to find C , however, since any point x on the hyperplane $x^T \lambda = W_F + W_B$ satisfies (11).

To estimate capacities, we take n measurements to find link prices as a function of different weights. To take measurement i , we pick some foreground flow i , increase its weight w_i slightly and hold all the other weights constant. This gives us a new sum of our weights $W_F^{(i)}$. Using (9), we find the set of link prices $\lambda^{(i)}$. After n measurements, we have the following matrix equation for $C_1, \dots, C_L$ and W_B ,

$$\begin{bmatrix} \lambda_1^{(1)} & \lambda_2^{(1)} & \dots & \lambda_L^{(1)} & -1 \\ \lambda_1^{(2)} & \lambda_2^{(2)} & \dots & \lambda_L^{(2)} & -1 \\ \vdots & & \ddots & & \vdots \\ \lambda_1^{(n)} & \lambda_2^{(n)} & \dots & \lambda_L^{(n)} & -1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_L \\ W_B \end{bmatrix} = \begin{bmatrix} W_F^{(1)} \\ W_F^{(2)} \\ \vdots \\ W_F^{(n)} \end{bmatrix}. \quad (12)$$

Note that if some link l isn't active in every measurement, we may not have enough data to estimate its capacity. To avoid this, we take an initial measurement and use (9) to find the set of prices. We find the set of bottlenecks by identifying which links have a nonzero price. We then ignore any subsequent measurement that has a different set of bottlenecks.

We have not been able to prove that this process produces independent row or column vectors. Experimentally, we generated twelve hundred random, identifiable networks and used (12) to estimate capacities. Given an initial measurement with k bottleneck links, we were always able to find full-rank matrix which can be solved exactly for the k bottleneck capacities and combined background weight. This takes an average of $1.5(k+1)$ measurements, due to some measurements resulting in a different set of bottlenecks.

6. REFERENCES

- [1] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, New York, NY, USA, 2004.
- [2] T. Bu, N. Duffield, F. L. Presti, and D. Towsley. Network tomography on general topologies. *SIGMETRICS Perform. Eval. Rev.*, 30(1):21–30, June 2002.
- [3] R. L. Carter and M. Crovella. Measuring Bottleneck Link Speed in Packet-Switched Networks. *Perform. Eval.*, 1996.
- [4] R. Castro, M. Coates, G. Liang, R. Nowak, and B. Yu. Network tomography: recent developments. *Statistical science*, pages 499–517, 2004.
- [5] C. Dovrolis, P. Ramanathan, and D. Moore. What Do Packet Dispersion Techniques Measure? *INFOCOM*, 2001.
- [6] F. P. Kelly, A. K. Maulloo, and D. K. Tan. Rate control for communication networks: shadow prices, proportional fairness and stability. *Journal of the Operational Research society*, 49(3):237–252, 1998.
- [7] R. Srikant and L. Ying. *Communication Networks: An Optimization, Control and Stochastic Networks Perspective*. Cambridge University Press, New York, NY, USA, 2014.
- [8] D. X. Wei, C. Jin, S. H. Low, and S. Hegde. Fast tcp: motivation, architecture, algorithms, performance. *IEEE/ACM Transactions on Networking (ToN)*, 14(6):1246–1259, 2006.